



COMP3153/9153

Algorithmic Verification

Lecture 18: Course summary

Final exam

- Available on the course website at **9:00 (AEST) on Friday, August 18**
- Submission via give/webCMS at **9:00 (AEST) on Saturday, August 19**
- I will be on ed to answer questions – any pertinent information will be posted on course website/ed/email
- Prose should be typeset
- Combination of short answer and long answer questions
- Review tutorial and practice questions, past exam paper

Materials

The questions are intended to assess your understanding, your ability and critical thinking skills, rather than your knowledge.

Unless specified, *any* approach is acceptable.

Access to *passive sources* (i.e. *existing* material) is permitted – should be properly referenced.

Accessing *active sources* (e.g. collaborating with other students, actively seeking answers through forums/social media, contract cheating, etc) is a breach of the University's Plagiarism Policy and will be subject to disciplinary action as outlined in the Student code of conduct

Answers must be typeset, though hand-drawn diagrams are ok.
Maximum size 15Mb.

NB

Questions are culturally neutral – no concepts other than those

Special consideration

Review the UNSW policy on Special Consideration.

UNSW has a “fit-to-sit” policy: by undertaking the assessment you are declaring that you are fit to do so.

If there are any foreseeable issues you must apply before the exam. If circumstances prevent you from applying before the exam, you must apply as soon as possible (within 3 days).

NB

Supplementary exams are only available to students granted Special Consideration.

Technical issues

If you experience technical issues before or during the exam, you should follow these instructions:

- Take screenshots (including time and date) of as many of the following as possible:
 - error messages
 - screen not loading
 - timestamped speed tests
 - power outage maps
 - messages or information from your internet provider regarding the issues experienced
- Make contact with me immediately (through email, ed, or webCMS) and advise me of the issue.
- Submit a Special Consideration application immediately at the conclusion of your assessment and upload your screenshots.

Assessment summary

- Formatif 50%, Exam 50%
- Formatif:
 - Existing tasks plus **one** more, complete at appropriate level.
Remaining tasks are optional.
- Exam: 40% barrier

Week 1 recap

- Automata
- LTL, CTL, CTL*
 - Syntax
 - Semantics

Week 2 recap

- Expressiveness of LTL, CTL, CTL*
- CTL model checking (marking algorithm)
- Symbolic model checking

Week 3 recap

Ordered Binary Decision Diagrams

- Representation of Boolean Functions
- Stores information in paths of a directed acyclic graph
- Construction:
 - Start with a binary decision tree
 - Merge identical subtrees
 - Remove redundant nodes

Safety, Liveness and Fairness

- Safety properties
 - Nothing bad happens
 - Property refuted by finite prefix
- Liveness properties
 - Good things eventually happen
 - Every finite prefix can be extended to an acceptable sequence

Week 4 recap

- Büchi automata
 - Union and intersection
 - Language emptiness testing
- LTL model checking
 - Automaton for Kripke structure
 - Local automaton
 - Eventuality automaton

SPIN and Promela:

- <http://www.spinroot.com>
- Code from lectures on course website (shortly)

Week 5 recap

Predicate abstraction

- CEGAR: Abstraction — (Spurious) counterexample — Refinement

Simulation and Bisimulation

Week 7 recap

Verification games

- Arenas, strategies
- Winning conditions: Büchi, Parity, Rabin, Muller
- Logic formula evaluation game
- Model checking games

Bounded Model Checking

- Reducing state space – what are the issues?
- LTL semantics: dealing with finite and infinite paths

Week 8 recap

Static analysis

- Fixed points
- Interval abstraction
- Data flow analysis:
 - Available Expressions (forwards must)
 - Live Variables (backwards may)

Timed Automata

Week 9 recap

- Timed logics: MTL, TPTL, TCTL, $TCTL_C$
- Expressiveness of temporal and timed logics

The end...

Questions/comments?

The end...

Questions/comments? Thank you!